

Qatar-1b

R-2770

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_140609 · created 2026-09-03T23:04:28+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456817.73264 +/- 0.00031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.022
Transit depth (Rp/Rs)^2	1.82 +/- 0.59 [%]
Orbital Inclination (inc)	88.39 +/- 1.84
Airmass coefficient 1 (a1)	1.02 +/- 0.019
Airmass coefficient 2 (a2)	-0.0115 +/- 0.0096
Scatter in the residuals of the lightcurve fit is	1.9 %
Best Comparison Star	#3 - [56, 290]
Optimal Method	PSF photometry
Transit Duration (day)	0.0801 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.135 ± 0.022 vs 0.1463 (deviation 0.51σ)
Duration fit vs published	0.0801 d vs 0.0692 d (deviation 3.12σ)
Mid-time O-C	-0.8 min
Depth SNR	6.1
Residual scatter	1.9 %

Light curve

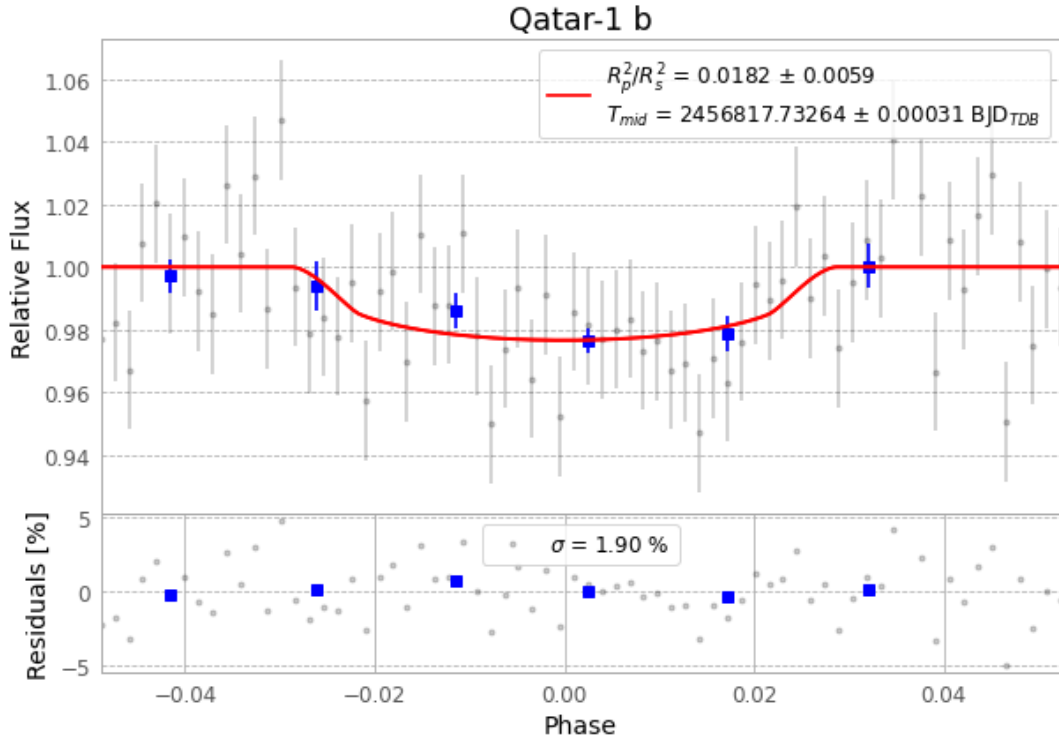


Figure 1 — FinalLightCurve_Qatar-1 b_2014-06-08.png (SHA-256 60102041cf1bf8a9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [324, 196], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a0ea0f56920ce1f0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

72 FITS frames (47 MB), Qatar-1140609034812.FITS → Qatar-1140609072112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-140609.log (e0d1a48d4914...), FinalLightCurve_Qatar-1 b_2014-06-08.png (60102041cf1b...), FinalLightCurve_Qatar-1 b_2014-06-08.csv (0e5b9f81c25d...)

Compute resources

Wall time 131.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 23:04Z.